POSTER

Reducing Barriers to Entry by Removing Prerequisites for a CS1 Introductory Programming Course

UDAYAN DAS, Saint Mary's College of California, Moraga, CA, United States

CHRIS FULTON, Loyola University Chicago, Chicago, IL, United States

Open Access Support provided by:

Loyola University Chicago

Saint Mary's College of California

Published: 15 March 2024

Citation in BibTeX format

SIGCSE 2024: The 55th ACM Technical Symposium on Computer Science Education
March 20 - 23, 2024
OR, Portland, USA

Conference Sponsors:
SIGCSE

SIGCSE 2024: Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2 (March 2024)
https://doi.org/10.1145/3626253.3635492
ISBN: 9798400704246

Reducing Barriers to Entry by Removing Prerequisites for a CS1 Introductory Programming Course

Udayan Das
Mathematics and Computer Science
Saint Mary's College of California
Moraga, CA
udd1@stmarys-ca.edu

Chris Fulton
School of Continuing and Professional Studies
Loyola University Chicago
Chicago, IL
cfulton@luc.edu

ABSTRACT

Introductory programming has evolved in many places to become a CS0 course, enabling students to get their feet wet with programming without completing significant math coursework. A scan of CS programs shows that a majority of CS1 programming courses that count towards an undergraduate CS degree continue to have a math or CS0 prerequisite. This experience report discusses the impact of removing the math prerequisite at an R2 university and a small liberal arts college. Having minimal prerequisites has beneficial effects in terms of diversifying the CS student body as well as enabling students to begin CS coursework early, often in the first semester, potentially impacting persistence, but also enabling students to decide, early, if CS is right for them. The high success rate of students of various backgrounds taking CS certificates and pursuing graduate school also shows that aggressive prerequisites in the past may have been functioning as barriers to entering CS programs. If we are serious about supporting diversity, we need to acknowledge the wide disparity in high school education nationwide and that prerequisites are perhaps functioning as a needless barrier. Where the CS0 course doesn't count towards a degree, or there isn't space for that requirement in the program, it is also worth considering whether the CS0 prerequisite is necessary.

KEYWORDS

CS1, Intro Programming, Diversity, Barrier reduction, Broadening Participation in Computing (BPC)

ACM Reference Format:
Udayan Das and Chris Fulton. 2024. Reducing Barriers to Entry by Removing Prerequisites for a CS1 Introductory Programming Course. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2 (SIGCSE 2024), March 20-23, 2024, Portland, OR, USA.* ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/3626253.3635492

SIGCSE 2024, March 20–23, 2024, Portland, OR, USA

ACM ISBN 979-8-4007-0424-6/24/03.
https://doi.org/10.1145/3626253.3635492

1 INTRODUCTION

Prior to Fall 2018, the CS1 introductory programming course at an R2 university had calculus as a prerequisite. The same was true of a similar course at a liberal arts college until Spring 2022. This poster details the experience of having removed the calculus prerequisite and requiring minimal prerequisites for the CS1 programming course. Author 1 was the program director at the R2 university when the change was put in place and has since moved to the liberal arts college as of Fall 2021. Author 2 is the current program director at the R2 who has left the minimal prerequisite requirement for the CS1 course as is based upon their prior experience with similar requirements at a large public R1 university and a technical college offering associate degrees. Several major types of academic institutions are thus covered by our collective experience, and this demonstrates that most students are prepared to study programming at a CS1 level right away provided there are appropriate teaching and learning methodologies applied. At the R2 and the liberal arts college the change to the prerequisite was coupled with peer tutoring; active learning; project-based learning; and strong student support through academic advising, course touchpoints, and student success coaches.

Evolution. There is an evolution in progress as far as what prerequisites are required to begin a CS1 programming course. In a survey scan, we found that 30% of institutions have limited or no prerequisites beyond high school or GED completion required for students starting a CS1. This no prerequisite approach is consistent with our experience and the purpose of this experience report is to share information with those programs that still have prerequisites, especially calculus prerequisites, and to encourage them to reconsider those requirements.

Impact. There has been a significant increase in students with no calculus background opting to take the course and when students are taking the course. Both authors have strongly encouraged students taking the CS1 course as early as possible and the now almost all students take the CS1 course in the first 2 semesters, with the vast majority taking it in the first semester. (At the R2 the semester is split into two 8-week terms.) Although this poster does not focus on graduate students from non-CS backgrounds, it is worth noting that at the R2 those students in the CS and CS-adjacent masters programs often take the CS1 course as prep for beginning graduate study. There, as in a majority of graduate CS and CS-adjacent programs there is no explicit undergraduate calculus requirement to begin graduate study in CS. This is a significant observation and worthy of reflection by all CS faculty as to the necessity of the prerequisites at the undergraduate level. We believe this also supports our observation that student motivation is the biggest factor in ensuring student success in the CS1 programming course.

CS1 programming topics. We use Tew et al. [1], curriculum 2013, and curriculum 2023 beta [2] as guides for a set of topics common to a CS1 programming course.

Table 1: CS1 topic areas

Variables
Data types
Console I/O
Expressions
Branching and selection
Loops
Functions
File I/O
Modules and libraries
Objects
Recursion

Scan results. We conducted a scan of CS programs based on publicly available information for 50 programs in the US. These programs included many top CS programs as well as academic institutions of various types, including Carnegie Classified R1 and R2 universities, private and state universities, liberal arts colleges, as well as community colleges.

Table 2: Scan Results of CS programs scan

CS1 course Prerequisite	Percentage
None	30%
Calculus or precalculus	24%
Calc. or precalc. Co-req	12%
College algebra or math reasoning	12%
CS0	30%

* Note: numbers do not add up to 100 because listed requirements sometimes overlap.

2 INSTRUCTOR REFLECTIONS

Instructor 1 reflection. Key points:

- Primarily adult learners at R2.
- Non-profit training experience with underserved individuals taught me that those with little or no background could be brought up to speed as far as programming is concerned with the right teaching and learning approaches bolstered by the appropriate student supports.
- Calculus caused hesitation for students starting at R2 and discouraged many from starting the program. Calculus was a daunting prospect for students returning after many years.
- Calculus prerequisite added a course to 3 course certificate.
- Incoming CS graduate students at R2 do not necessarily need calculus and CS department began having those students take course at the continuing studies unit.
- At liberal arts college there was also calculus requirement which was removed. The BA at liberal arts college now does not require calculus and that has encouraged students majoring in English and Anthropology to double major in CS, which would be harder with the stringent math requirements.
- Students from other departments can also pursue 3 course CS certificate or CS minor without the math requirement.
- I think the greatest driver of student success in CS1 is motivation and engagement.
- Bringing in wider preparation also brings wider background of students, enriching overall learning.

Instructor 2 reflection. Key points:

- An advocate for excluding calculus as an entry criterion for CS1 while not eliminating the course.
- Immediate engagement in technical skills motivated students to continue in coursework.
- Course accessibility to non-computer science majors broadened its reach and value for different majors.
- Students from other disciplines found an interest in programming and continued in subsequent courses.
- No prerequisites for CS 1 made it easier for administrators to approve enrollment requests as a general elective if space was available.
- The requirement of prerequisites, especially math courses, deterred students from enrolling in CS1.
- I observed that students at various types of institutions could succeed in CS1 without traditional prerequisites.
- Removing prerequisites for CS1 can attract more returning students without imposing math-related barriers. Adult learners face higher barriers to returning to school compared to traditional students transitioning from high school and a prerequisite can be a barrier for those whose interest is focused on learning the skill of programming.

3 CONCLUSIONS

In our experience, removing all prerequisites but high school completion or GED has had no impact on student performance in the introductory programming CS1 course. The removal of prerequisites has significant benefits in terms of allowing a wider variety of students to take the CS1 course earlier in their programs. The rationale of requiring students to take calculus prior to taking a CS1 course should be reconsidered. Not only is there no dependence of a standard CS1 programming course on calculus, but this requirement can also function as discouragement and a delay in students interested in taking the CS1 programming course. Towards broadening participation in computing and supporting students from a wide variety of backgrounds, bringing differing prior experience a big step would be making it easier for students to begin programming as soon as possible.

REFERENCES

[1] A. E. Tew and M. Guzdial, "Developing a validated assessment of fundamental CS1 concepts," in *Proceedings of the 41st ACM technical symposium on Computer science education*, Milwaukee Wisconsin USA: ACM, Mar. 2010, pp. 97–101. doi: 10.1145/1734263.1734297.

[2] "CS2023 – ACM/IEEE-CS/AAAI Computer Science Curricula." Accessed: Aug. 18, 2023. [Online]. Available: https://csed.acm.org/